

Problem Set 2

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Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.¹ As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, “We can solve this the easy way” to draw a bribe). The table below shows the resulting data.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

¹Fried, Lagunes, and Venkataramani (2010). “Corruption and Inequality at the Crossroad: A Multi-method Study of Bribery and Discrimination in Latin America. *Latin American Research Review*. 45 (1): 76-97.

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

- A Chi-squared test looks to determine if two categorical variables are statistically independent from each other. In this case we want to discern whether class and police response from traffic violation are statistically independent
- The null hypothesis H_O is that the variables - Class and Encounter - are statistically independent. Our alternative H_a is that they are not independent.
- This is determined by looking at the observed and expected frequencies of the data. The expected frequency is the frequency that we would observe if the null was true and the variables were independent.
- Our null therefore is that expected frequencies do not differ from observed frequencies. $H_O : f_O = f_e$, and if there is not a statistical significant difference between them we have evidence supporting this hypothesis.
- Alternatively, if we do find that our expected frequencies differ significantly from observed, $H_O : f_O \neq f_e$, we have evidence to reject the null and accept that the variables are not statistically independent.

First step is to load in data. After transferring the table data, add on a 'Total' row and column that sum up the total frequencies in each category respectively.

```
1 data <- matrix(c(14,7,6,7,7,1), nrow = 2)
2 colnames(data) = c("Not_Stopped", "Bribe_Requested", "Stopped_Warned")
3 rownames(data) = c("Upper Class", "Lower Class")
4 fO <- as.table(data) #frequency observed table
5 fO_t <- rbind(fO, Total = colSums(fO)) #add column total
6 fO_t <- cbind(fO_t, Total = rowSums(fO_t)) #add row total
7 print(fO_t) # table of observed frequencies with totals
```

Observed frequency with Totals

	Not_Stopped	Bribe_Requested	Stopped_Warned	Total
Upper Class	14	6	7	27
Lower Class	7	7	1	15
Total	21	13	8	42

We also determine the expected frequencies of the data, which are the frequencies expected if the categories had no impact on the other variable.

$$f_e = \frac{\text{row total}}{\text{grand total}} \times \text{column total} \quad (1)$$

```

1 row_totals <- rowSums(fO)
2 col_totals <- colSums(fO)
3 grand_total <- sum(row_totals)
4 fe <- outer(row_totals, col_totals, "*") / grand_total
5 fe <- round(fe, 2)
6 fe

```

Expected Frequencies

	Not_Stopped	Bribe_Requested	Stopped_Warned
Upper Class	13.5	8.36	5.14
Lower Class	7.5	4.64	2.86

Now we compare the observed and expected frequency tables to calculate a Chi-square.

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e} \quad (2)$$

```

1 res <- (fO - fe)^2 / (fe) #squared diff btw obs and exp values div by exp
2 chi_square <- sum(res) # sum across all frequencies
3 chi_square

```

```
[1] 3.801141
```

The overall test statistic of the Chi-squared is $\chi^2 = 3.80$

- (b) Now calculate the p-value from the test statistic you just created (in R).² What do you conclude if $\alpha = 0.1$?

Calculate the p-value, i.e the probability of finding our Chi-square value under the null hypothesis given the degrees of freedom.

```

1 df <- (nrow(fO)-1) * (ncol(fO)-1)
2 p_value <- pchisq(chi_square, 2, lower.tail = FALSE)
3 p_value

```

```
[1] 0.1494833
```

Since the p-value $p = 0.150$ is greater than set confidence level - $p = 0.150 > 0.1 = \alpha$ - we accept our null hypothesis that the two variables are statistically independent. This means we do not have evidence to accept the alternative hypothesis that class impacts the encounter when breaking traffic laws.

We can check these results with the built-in Chi-square function in R and see they are the same (bar rounding) to our by-hand values.

²Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

```

1 chisq.test(fO, fe)

Pearson's Chi-squared test
data:  fO and fe
X-squared = 3.7912, df =
2, p-value = 0.1502

```

(c) Calculate the standardized residuals for each cell and put them in the table below.

By calculating standardized residuals we can see how much each observed value differs from the expected. This is the difference between the observed and expected frequency adjusted by the standard error of the difference under the null hypothesis.

$$z = \frac{f_O - f_e}{se} = \frac{f_O - f_e}{\sqrt{f_e(1 - rowprop.)(1 - colprop.)}} \quad (3)$$

```

1 df <- (nrow(fO)-1) * (ncol(fO)-1)
2 p_value <- pchisq(chi_square, 2, lower.tail = FALSE)
3 p_value

```

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.32	-1.52	1.65
Lower class	-0.27	1.93	-1.52

(d) How might the standardized residuals help you interpret the results?

- Table is good to quickly visualise where the biggest differences are between observed and expected frequency. This may help find out where the largest residuals are that contribute to the chi-square.
- Cells also have different numbers of counts and therefore those with more counts are likely to have larger residuals overall, but by standardizing them, each cell's adjusted residuals can more fairly be compared.
- Another benefit of standardised residuals is they because they are standardised they approximate to the standard normal distribution $N(0,1)$, therefore can be used to conduct post-hoc tests on specific cells.
- The absolute value of the cells can be compared with critical values i.e $N(0,1)$ $1 - \alpha/2 = 1.96$ at $\alpha = 0.05$ level such that any standardized residuals greater than this level will be significant.
- Here we can see that none of the cells values have an absolute value equal to or greater than 1.96 indicating there is not a significant difference between expected and observed to a level $\alpha = 0.05$. This lines up with our overall Chi-squared result but if we were to find statistical independence this would allow us to see precisely where the variations came from.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.³ Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: <https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv>

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

```
1 GP <- read.csv("https://raw.githubusercontent.com/kosukeimai/qss/master/
  PREDICTION/women.csv", header=T)
```

³Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*. 72 (5), 1409-1443.

- (a) State a null and alternative (two-tailed) hypothesis.

Null hypothesis: Having a reservation policy has no effect on the number of new or repaired drinking-water facilities in the villages
Alternative hypothesis: Having a reservation policy has an effect on the number of new or repaired drinking-water facilities in the villages

- (b) Run a bivariate regression to test this hypothesis in R (include your code!).

```
1 reg1 <- lm(data= GP, water ~ reserved)
2 output_stargazer <- function(outputFile, ...) {
3   output <- capture.output(stargazer(...))
4   cat(paste(output, collapse = "\n"), "\n", file=outputFile, append=TRUE)
5 }
6 output_stargazer("PS02_reg1output.tex", reg1)

1 pdf("PS02_Reg_Plot.pdf")
2 plot(GP$reserved,
3      GP$water,
4      xaxt = "n",
5      xlab="Reserved Policy",
6      ylab="Number water facilities")
7 abline(reg1, col = "red") # Add regression line
8 axis(1, at = c(0,1))
9 dev.off()
```

Table 1:

	<i>Dependent variable:</i>
	water
reserved	9.252** (3.948)
Constant	14.738*** (2.286)
Observations	322
R ²	0.017
Adjusted R ²	0.014
Residual Std. Error	33.446 (df = 320)
F Statistic	5.493** (df = 1; 320)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Table 2:

	<i>Dependent variable:</i>
	water
reserved	9.252** (3.948)
Constant	14.738*** (2.286)
Observations	322
R ²	0.017
Adjusted R ²	0.014
Residual Std. Error	33.446 (df = 320)
F Statistic	5.493** (df = 1; 320)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

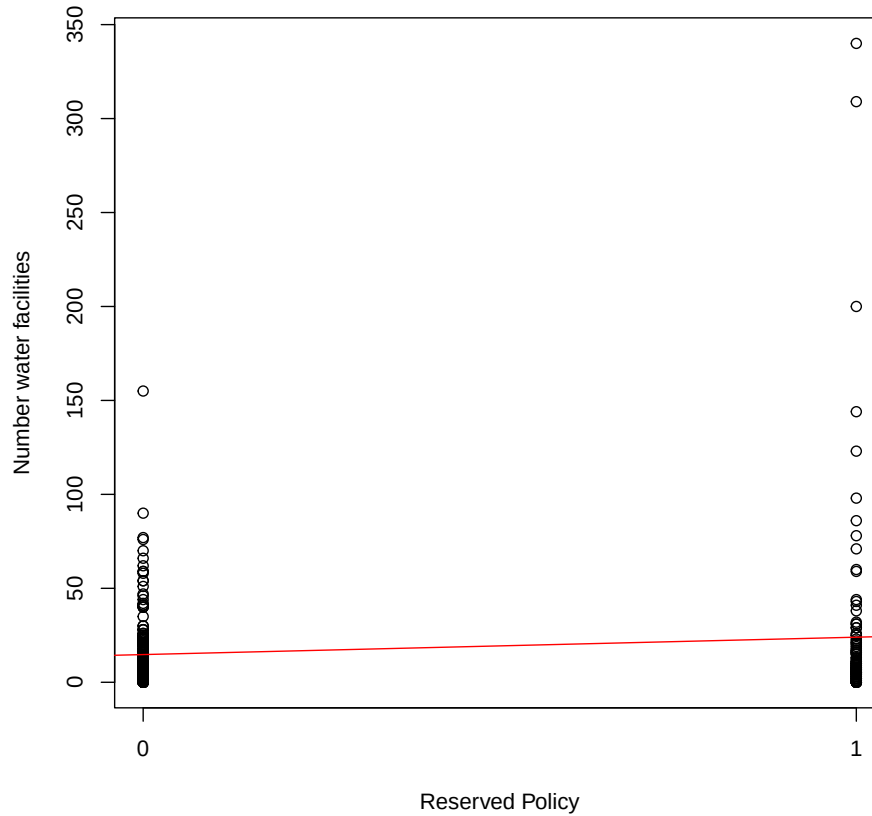
Table 3:

	<i>Dependent variable:</i>
	water
reserved	9.252** (3.948)
Constant	14.738*** (2.286)
Observations	322
R ²	0.017
Adjusted R ²	0.014
Residual Std. Error	33.446 (df = 320)
F Statistic	5.493** (df = 1; 320)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Table 4:

	<i>Dependent variable:</i>
	water
reserved	9.252** (3.948)
Constant	14.738*** (2.286)
Observations	322
R ²	0.017
Adjusted R ²	0.014
Residual Std. Error	33.446 (df = 320)
F Statistic	5.493** (df = 1; 320)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Figure 2: Plot of Water and Reserved Policy



(c) Interpret the coefficient estimate for reservation policy.

The coefficient estimate is the value from the model that shows how a change in one variable is associated with another, in this case looking at reservation policy on water facilities. The coefficient estimate $\hat{\beta}_1$ for reservation policy is 9.252. This means that having a reservation policy is associated with villages having 9.252 more new or repaired drinking-water facilities. The standard error of the estimate $se(\hat{\beta}_1) = 3.948$. At $\alpha = .05$ we are able to reject the null hypothesis $p < 0.05$ and accept that having a reservation policy does have a statistically significant impact on the number of new or repaired drinking-water facilities in a village.

We can constructing a 95% confidence interval

Confidence interval It is nevertheless important to remember the assumptions underpinning this interpretation, such as the relationship being linear, independent observations, normal and constant variance with normal errors. There is also other factors to consider, including

```
1 confint(reg1, level = 0.95)
```

	2.5 %	97.5 %
(Intercept)	10.240240	19.23640
reserved	1.485608	17.01924

Thus a 95% confidence interval of the estimate with $t_n - 2$ is $[1.49, 17.02]$ which does not include zero, further confirming our evidence that the estimate found is not 0. high correlation between reserved and women spurious / chain relationships discuss